



Do information releases increase or decrease information asymmetry? New evidence from analyst forecast announcements[☆]



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ABSTRACT

Prior literature documents that both earnings announcements and management earnings forecasts increase information asymmetry at announcement. In contrast, we predict and document that analyst earnings forecasts decrease information asymmetry at announcement. As expected, this directional contrast is temporary, in that all three information release types lead to a decrease in information asymmetry following the short-window announcement period. Our evidence demonstrates that the direction of the effect of a public information release on announcement-period information asymmetry is determined by how the information contained in the release relates to prior information held by sophisticated and unsophisticated investors, which supports extant disclosure theory.

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1. Introduction

Information asymmetry between "sophisticated" and "unsophisticated" investors (hereafter referred to simply as "information asymmetry") is a pervasive friction in capital markets (e.g. Amihud and Mendelson, 1986; Easley and O'hara, 2004; Kim and Verrecchia, 1994; Leuz and Verrecchia, 2000).¹ In this study, we revisit a fundamental question in

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¹ Similar to Kalay (2015), we use the term "sophisticated investors" in a broad sense. This broad definition of sophistication is meant to capture an investor's ability to utilize disclosed information in order to gain an information advantage. It describes active information users who devote more time and attention to their investments and are more proficient in analyzing investment-related information. Differences in sophistication can arise from heterogeneity in investors' opportunity cost of time and in their ability to acquire and analyze information. Disclosure theory establishes the idea that a group of more sophisticated investors may be apt to learn more from public information (e.g., Bushman et al., 1996; Fischer and Verrecchia, 1999; Kim and

information economics, which has implications for both accounting and finance—what effect does the public release of firm-specific information into the market environment (hereafter referred to as a “information release”) have on information asymmetry in a short window around the announcement event?

Whereas public information releases of all types generally decrease information asymmetry in the post-announcement period relative to the pre-announcement period, disclosure literature theorizes that the net effect of a given information release on information asymmetry *at announcement* depends on the relative strengths of two countervailing forces that exist simultaneously in any given information release (e.g., Kim and Verrecchia, 1994). First, an information release provides some information that is new to *both* sophisticated and unsophisticated investors, which will *increase* information asymmetry at announcement, *ceteris paribus* (hereafter referred to as the “asymmetry-increasing force”). This immediate increase occurs because of the superior processing abilities possessed by sophisticated investors relative to unsophisticated investors, which allows sophisticated investors to impound the information contained in the announcement more quickly than unsophisticated investors. Second, at the same time, an information release reveals to unsophisticated investors some information that is new to them but which was already known (privately) by sophisticated investors, which will *decrease* information asymmetry at announcement by reducing the information gap between the two types of investors, *ceteris paribus* (hereafter referred to as the “asymmetry-decreasing force”). Hence, in theory, the announcement effect of a public information release on information asymmetry depends on which type of information dominates in that particular release.

Relying on these theoretical arguments as motivation, prior literature finds that earnings announcements and management forecasts *increase* information asymmetry at announcement (e.g., Coller and Yohn, 1997; Lee et al., 1993). These findings are consistent with our intuition that, as firm-supplied information releases, earnings announcements and management forecasts are likely to contain more information that is new to all external market participants (relative to information that is new only to unsophisticated investors), resulting in dominance of the asymmetry-increasing force. Extant empirical literature has to our knowledge provided no evidence of an information release that decreases information asymmetry at announcement. Thus, there is no empirical evidence that the asymmetry-decreasing force plays a significant role in announcement-period information dynamics.

In this study, we utilize the setting of analyst forecasts to re-examine these dynamics. Compared to earnings announcements and management forecasts, extant literature suggests that analyst forecasts contain a relatively high concentration of information that is new only to unsophisticated investors (i.e., already known by sophisticated investors) versus information that is new to all (e.g., Irvine et al., 2007; Kadan et al., 2016). Conceptually, an analyst forecast is an information release by a sophisticated information intermediary who collects public and private information and analyzes it, which parallels the activities of sophisticated investors. Indeed, financial analysts are often thought of as a proxy for sophisticated investors in capital markets. As an example, one may think about a sophisticated investor as an institution (e.g., hedge fund) that employs buy-side analysts. The hedge fund's buy-side analysts have similar skill sets and industry connections, and perform similar analysis, information collection and research as do sell-side analysts, and at certain times collect information directly from sell-side analysts.² Therefore, it is likely that the information set possessed by sophisticated investors will have relatively high overlap with the information contained in analyst forecasts. Based on this logic, we predict and document that analyst earnings forecasts decrease information asymmetry at announcement.

Of course, analyst forecasts may indeed contain some information that is new to both sophisticated and unsophisticated investors, which would produce the asymmetry-increasing force. However, our study is based on an argument of relativity. That is, whereas the asymmetry-increasing force is dominant in earnings announcements and management forecasts, we expect the asymmetry-decreasing force to be stronger in analyst forecasts, relatively speaking. This logic is further supported by extant evidence that suggests that the information set of analysts overlaps with the information set of sophisticated investors, and by evidence of investor trading patterns around analyst forecast announcements.

Although analysts produce several types of information releases (e.g., earnings forecasts, stock recommendations, target prices), we focus our attention on analyst earnings forecasts, as these releases are the most frequent type of output, and are more directly comparable with earnings announcements and management earnings forecasts. The three types of events are comparable in that they are frequently recurring releases of information events that are well-studied (i.e., we understand most of their properties), where the type of information is similar in nature (i.e., corporate earnings/performance). Nevertheless, we acknowledge that there are characteristic differences between analyst forecasts and both earnings announcements and management forecasts, apart from the theoretical difference that we rely on to generate our prediction (i.e., that analyst forecasts contain a higher relative concentration of information that is new only to unsophisticated investors). For example, analyst forecasts may less likely be anticipated events, relative to earnings announcements and management forecasts. As another example, extant literature suggests that analyst forecasts may contain differential biases, relative to earnings announcements and management forecasts (e.g., McNichols and O'Brien, 1997). In our empirical tests,

(footnote continued)

Verrecchia, 1994). Therefore, the economic idea that some investors benefit relatively more from certain disclosure activities is an extension of this literature, and can relate to differences in investors' ability to utilize disclosed information (e.g., Bloomfield, 2002; Hirshleifer and Teoh, 2003).

² We use hedge funds here for illustration only. There are also small sophisticated investors that can perform similar research tasks on a smaller number of firms. Also, one may think about many types of unsophisticated investors, including low-cost mutual funds that do not employ buy-side analysts, and casual investors.

we address these factors to increase the confidence that our directionally opposite results across information release types are driven by the differential information dynamics that we theorize.

We follow prior literature and focus on equity market bid-ask spread as a measure of information asymmetry, while controlling for its non-information-asymmetry-related components. We document that, on average, bid-ask spread decreases by approximately one-and-a-half percent during an analyst forecast announcement period relative to the period immediately preceding the forecast. Because there are, on average, between two and three analyst forecasts per firm-quarter, the overall quarterly magnitude of the real economic effect of analyst forecasts on a given firm's bid-ask spread during the announcement period is therefore approximately four percent. We find consistent results using quoted depth and price impact as alternative measures of information asymmetry.

Although our paper focuses on announcement-period effects, we further document that all three types of information releases (i.e., analyst forecasts, management forecasts and earnings announcements) decrease information asymmetry over the ten days subsequent to the announcement. Accordingly, consistent with our expectations, the directionally opposite information asymmetry effects across analyst forecasts and the other two information releases are temporary, and exist only during the short-window announcement period.

We conduct several additional analyses to provide comfort that our findings are not attributable to alternative explanations. First, we provide evidence that our results hold in a subset of forecasts that are isolated in time relative to other analyst forecasts, which suggests that our findings are not attributable to the alternative explanation that analysts simply issue their forecasts in the wake of other significant information events while information asymmetry is already decreasing. Second, we show that the magnitude of the information asymmetry decrease is a function of forecast information content, which is difficult to reconcile with alternative explanations. Third, we document that our results hold for both “anticipated” and “unanticipated” analyst forecasts, which suggests that announcement anticipation is not the driver of our findings. Finally, our results hold for both good news and bad news forecasts. This mitigates the concern that our results may be attributable to analysts' incentives to disclose good news, or to a mechanical relation between the nature of news and bid-ask spread.

Our evidence contributes to the literature by documenting that different types of information releases lead to predictably opposite effects on information asymmetry at announcement. Specifically, our evidence that analyst forecasts cause reductions in announcement-period information asymmetry in equity markets complements the evidence provided in [Coller and Yohn \(1997\)](#) and [Lee et al. \(1993\)](#) which show that earnings announcements and management forecasts increase announcement-period information asymmetry. Although these directionally opposite effects are temporary, the understanding of how information is incorporated into prices and short-term trading frictions is central to our understanding of capital markets phenomena, and requires the study of short-window capital market dynamics (e.g., [So and Wang, 2014](#)). Relatedly, these results also provide the first empirical validation (of which we are aware) of the discussion in [Kim and Verrecchia \(1994\)](#) that an information release can indeed decrease information asymmetry at announcement.

In addition, our study speaks to the contribution of analysts to a firm's information environment. Specifically, our results suggest that on average, analyst forecasts lead to an immediate narrowing of the information gap between sophisticated and unsophisticated investors. Further, our short-window announcement-period event study design reduces concerns relating to reverse causality that have plagued prior literature (e.g., [Healy et al., 1999](#); [Lang and Lundholm, 1996](#)), and therefore contributes to the recent debate about the causal effects of sell-side equity research ([Kelly and Ljungqvist, 2012](#); [Li and You, 2015](#)).

2. Motivation and predictions

Extant literature characterizes information asymmetry between sophisticated and unsophisticated investors (e.g., [Glosten and Milgrom, 1985](#); [Kalay, 2015](#)). Typically, such information asymmetry is thought to come from two potential sources. First, sophisticated investors have access to (or the ability to develop) private information about the firm that is not available to unsophisticated investors. Second, sophisticated investors have superior processing abilities with respect to new information when it is released to all market participants, which allows sophisticated investors to gain a temporary information advantage with respect to new public information.

Consistent with these dual sources of information asymmetry, [Kim and Verrecchia \(1994\)](#) observe that the announcement-period effect of a public information release on information asymmetry can be characterized by two simultaneous forces that have directionally opposite empirical implications.³ First, an information release that provides new information to both sophisticated and unsophisticated investors *increases* information asymmetry at announcement, because sophisticated investors can process and incorporate information more quickly than unsophisticated investors. Second, at the same time, an information release *decreases* information asymmetry at announcement if it reveals to unsophisticated investors information that they did not previously possess but which the sophisticated investors previously held privately. The net directional effect on announcement-period information asymmetry therefore depends on how the information contained in the release relates to the information previously held by sophisticated investors.

³ [Kim and Verrecchia \(1994\)](#) consider many financial information releases to fit this characterization, including earnings announcements, management forecasts, and analyst forecasts.

Extant literature provides evidence consistent with dominance of the asymmetry-increasing force. Lee et al. (1993) find that bid-ask spreads increase at the time of earnings announcements. Similarly, Krinsky and Lee (1996) provide evidence that the adverse selection component of the bid-ask spread increases significantly around earnings announcements, supporting the notion that earnings announcements indeed increase announcement-period information asymmetry. Similar to the evidence from earnings announcements, Collier and Yohn (1997) find that bid-ask spread increases in the short window around the issuance of a management forecast. Following the theory in Kim and Verrecchia (1994), these results suggest that, consistent with our intuition, earnings announcements and management forecasts contain a relatively high proportion of information that is new to both sophisticated and unsophisticated investors (although, of course, we expect that they also contain some information that is already known to sophisticated investors but new to unsophisticated investors). These findings are further consistent with the notion that order flow immediately after earnings announcements and management forecasts is likely to come from sophisticated investors because of their superior ability to process new information. Intuitively, extant studies also find that this announcement-period increase in information asymmetry is temporary, in that bid-ask spreads one week after the management forecast announcement are smaller than spreads one week prior to the announcement.

Analyst forecasts have distinguishing characteristics that may make the asymmetry-decreasing force more pronounced, relative to earnings announcements and management forecasts. Specifically, an analyst forecast is an information release by a sophisticated information intermediary who collects public and private information and analyzes it, which parallels the activities of sophisticated investors. Indeed, financial analysts are often considered to be a proxy for sophisticated investors in capital markets (e.g., Allee et al., 2007; Ramnath et al., 2008). It is likely, then, that compared to earnings announcements and management forecasts, a relatively large proportion of the information contained in an analyst forecast is new only to unsophisticated investors, thus decreasing information asymmetry with sophisticated investors. This intuition yields our prediction that, whereas both earnings announcements and management forecasts increase information asymmetry at announcement, analyst forecasts decrease information asymmetry at announcement. Intuitively, when a specialist observes order flow immediately after an analyst forecast, she is less concerned than she was immediately before the analyst forecast release that the order flow is coming from a sophisticated investor holding an information advantage, and bid-ask spread should decrease accordingly.

Prior research supports this assertion that analyst forecasts have a high degree of overlap with information held by sophisticated investors. For example, extant evidence suggests that sophisticated investors place relatively little value on the earnings forecasting role of analysts. Brown et al. (2015) survey buy-side analysts, and document that only 11% view analyst forecasts as useful. Groysberg et al. (2015) find that broker votes for analysts, which determine their compensation, have very little relation to analyst forecast activity, and very strong relation with analysts' white papers (comprehensive research reports). Moreover, the literature on analyst tipping suggests that analysts directly reveal their information to sophisticated investors prior to public release (e.g., Irvine et al., 2007; Juergens and Lindsey, 2009; Markov et al., 2011), consistent with literature which shows that sophisticated investors can predict the content of analyst information releases prior to public announcement (e.g., Christophe et al., 2010; Klein et al., 2014). Further, a recent paper by Kadan et al. (2016) shows that institutions trade before analysts' information releases, while individuals trade mainly during and after the information releases. Consistent with our arguments above, these trading dynamics around analyst information releases are opposite to those documented in Lee (1992) for earnings announcements.

We also note that, whereas prior literature examines how the presence of information intermediaries affects the increase in information asymmetry surrounding earnings announcements, there is no prior evidence on the direct effect of analyst forecast releases on information asymmetry. Although Yohn (1998) finds evidence that general information asymmetry is lower for firms with greater analyst following during the period leading up to and including earnings announcements, she finds no evidence that the presence of analysts mitigates the earnings announcement increase in information asymmetry. Bushee et al. (2010) provide evidence that greater press coverage is associated with a lower abnormal bid-ask spread around earnings announcements. Whereas these studies examine interactive effects between the presence of information intermediaries and the effect of earnings announcements on information asymmetry, our study differs in that we examine the direct effect of information releases by analysts on information asymmetry, where by design we eliminate associated earnings announcement effects.

3. Research design and sample selection

3.1. General empirical setup

To test our predictions concerning the effect of an information release on announcement-period information asymmetry, we directly measure how the information asymmetry component of bid-ask spread changes across several days spanning the announcement date.⁴ One way to conceptualize this design choice is that as a specialist observes order flow, she increases bid-ask spread to protect herself when it is more likely that order flow is coming from sophisticated investors that have superior

⁴ We follow the bulk of extant literature and use equity bid-ask spread as our primary measure of information asymmetry.

processing abilities.⁵ However, it is well established that observed bid-ask spreads reflect not only information asymmetry, but also specialists' order processing costs and inventory carrying costs. To control for these other components of bid-ask spread, we estimate a model based upon finance theory (e.g., [Glosten and Milgrom, 1985](#); [Stoll, 1978](#)), variations of which have been empirically implemented in extant literature (e.g., [Amiram et al., 2011](#); [Coller and Yohn, 1997](#)). Specifically, we estimate the following model, which uses daily bid-ask spread as the dependent variable and several independent variables to absorb its non-information asymmetry components, where we use four daily observations spanning each announcement date (i.e., the two days before the announcement date, the announcement date itself, and one day after the announcement):

$$\begin{aligned} BASpread_{i,d} = & \beta_0 + \beta_1 PreInfo1_{i,d}^{IR} + \beta_2 Info0_{i,d}^{IR} + \beta_3 Info1_{i,d}^{IR} + \beta_4 Price_{i,d} \\ & + \beta_5 LnSize_{i,t-1} + \beta_6 Volatility_{i,t-1} + \beta_7 Turnover_{i,t-1} + \beta_8 Depth_{i,d} \\ & + \beta_9 Volume_{i,d} + \beta_{10} AbsCAR_{i,r} + \epsilon_{i,d}, \end{aligned} \quad (1)$$

where variable subscripts i , d , t , and r reference firm, day, quarter, and a particular information release, respectively. $BASpread$ is bid-ask spread percentage on trading day d , measured from TAQ.⁶ Our primary variable of interest is $Info0^{IR}$, which is an indicator variable that equals one if day d is the announcement day for an information release concerning firm i , and equals zero otherwise. We also include $Info1^{IR}$, an indicator variable that equals one if day d is the trading day immediately after the announcement day, as announcement-period effects may extend into a second day. $PreInfo1^{IR}$ is an indicator variable that equals one if day d is the trading day immediately prior to the announcement day. In this empirical design, bid-ask spread two days prior to the announcement day is reflected in the intercept (implicitly $PreInfo2^{IR}$), thus the included indicator variables $PreInfo1^{IR}$, $Info0^{IR}$, and $Info1^{IR}$ capture the change in bid-ask spread on those days relative to two days prior to the information release. The information release can be one of three types: an analyst forecast ($IR=AF$), earnings announcement ($IR=EA$), or management forecast ($IR=MF$). If analyst forecasts indeed decrease announcement-window information asymmetry as we predict, we expect $\beta_2 < 0$ and $\beta_3 < 0$ (where $IR=AF$).

We include daily stock price ($Price$) to control for market makers' processing costs ([Stoll, 1978](#)), prior-quarter ($t-1$) average daily turnover ($Turnover$) to control for liquidity of the firm's shares which affects inventory holding costs ([Demsetz, 1968](#)), and prior-quarter average return volatility ($Volatility$) and firm size ($LnSize$) to control for inventory risk.⁷ We include daily quoted depth ($Depth$) to control for the market maker's potential adjustment to depth as an alternative protection mechanism against inventory risk or information asymmetry. Further, we include daily trading volume ($Volume$) and the absolute value of the announcement window abnormal return ($AbsCAR$) to further control for inventory risk and differential news content. We also consider specifications that use quoted depth ($Depth$) and price impact of trades ($Pricelmpact$) as alternative dependent variables, as prior literature nominates these constructs as alternative measures of information asymmetry. We define all variables further in the [Appendix A](#).

Because the two firm-days immediately prior to the announcement day serve as a baseline for measuring announcement-window effects, endogeneity concerns that plague prior studies which examine the relation between analysts and a firm's information environment are mitigated.⁸ We introduce several modifications to this basic empirical structure in subsequent analyses. We discuss these modifications along with the associated results in [Section 4](#).

3.2. Sample selection and data

We obtain analyst forecast data from the I/B/E/S detail EPS US file, and restrict our sample to one-quarter-ahead forecasts (i.e., $fpi=6$) between the years 1993 and 2012, inclusive. We require observations to have non-missing data for the forecast announcement date ($anndats$), forecast value ($value$), actual quarterly earnings ($actual$), earnings announcement dates ($anndats_act$) company identifier ($ticker$), forecast period ending date ($fpedats$) and the forecasting analyst ($analys$).⁹ This yields a sample of 2,274,742 observations. Our analyses require stock price data from CRSP and quotes data from TAQ. Accordingly, we delete any observation for which we cannot obtain both a CRSP *permno* and TAQ *symbol*. This reduces the sample size to 2,195,458 observations. Further, we delete observations where the forecast date occurs after the earnings announcement date, as these represent data errors. Next, we remove analyst forecasts that are within plus or minus two days of an earnings announcement or management forecast by the forecasted firm in order to decrease any potential

⁵ A similar conceptualization exists for markets without specialists, as unsophisticated investors directly assess the likelihood that they are trading against sophisticated investors, and price protect accordingly.

⁶ Inferences that we report in subsequent analyses are unchanged if we use unscaled bid-ask spread, or scale by natural log of total assets rather than price.

⁷ Inferences are unchanged throughout our analyses if we add interactions between *Price* and $Info0,1^{IR}$ to further control for changes in price during the announcement period.

⁸ In untabulated analyses we compare the announcement-period effects with different benchmark periods before the announcement (in addition to the two days immediately preceding the announcement). Inferences from this alternative design are unchanged compared to our reported design.

⁹ We delete observations where *analys*=000000, as that code indicates that the particular analysts associated with a given forecast is unidentified.

confounding effects, where we obtain earnings announcement and management forecast data from Compustat and First Call, respectively.¹⁰ As a consequence, the sample size decreases further to 1,250,911 observations.

Next, we delete analyst firm-forecast-day observations if there is another analyst forecast for the same firm in the subsequent or preceding two days to prevent overlapping observations that may confound our measurement of the announcement window. This reduces the sample to 646,929 observations. Then, from CRSP we obtain prior-quarter return volatility and prior-quarter average turnover, as well as the absolute value of cumulative abnormal returns during the two-day announcement period. We next append daily TAQ bid-ask spreads and quoted depth, and daily CRSP stock prices and trading volume for the four-day window around each forecast announcement day (i.e., beginning two days prior to the forecast). In addition, we winsorize all continuous variables at the 1% and 99% levels, and only retain observations that have non-missing data for the complete four-day window. This results in a sample of 2,410,720 daily observations, which are by construction based on 602,680 distinct firm-forecast-day observations. Note that each observation is a unique combination of firm, fiscal period end, forecasting analyst, forecast date, and trading day relative to the forecast.

Our unit of study is a day on which a forecast is issued for a given firm. Accordingly, we only desire one observation for a firm-forecast day, regardless how many different individual analysts issue a forecast on that day. Therefore, we retain only one observation for each day on which a forecast was issued for a given firm-quarter. Consequently, our final sample contains 2,158,116 daily observations, which relate to 539,529 analyst forecasts (i.e., the difference between 2,410,720 and 2,158,116 reflects the fact that more than one analyst may issue a forecast for the same firm on the same day). Hereafter, we refer to this as our “full sample”.

Panel A of Table 1 presents descriptive statistics for our full sample. Unconditional bid-ask spread is 86.83 basis points, on average. The average bid-ask spread is lower than the bid-ask spreads in Lee et al. (1993) and Coller and Yohn (1997) due to the inclusion of more recent years in our sample period, which are characterized by improved market liquidity and lower bid-ask spreads relative to earlier sample years. Our sample firms are relatively large, with mean market value of equity of \$1.4 billion. Panel B of Table 1 presents a correlation matrix, with Pearson (Spearman) correlations reported above (below) the diagonal. Consistent with extant literature and models of bid-ask spread determinants, there are significant correlations between bid-ask spread and all of our control variables.

4. Empirical results

In this section, we present and discuss our empirical inferences. In our initial analyses, we report regression statistics based on several alternative configurations of fixed effects and clustered standard error structures, to demonstrate that our inferences are not sensitive to these choices. These configurations include (1) year fixed effects, with standard errors clustered by firm, which is a method recommended by Petersen (2009) to address both time-series and cross-sectional error correlations when the number of sample years is insufficient for clustering; (2) firm fixed effects, with standard errors clustered by both firm and year-quarter (e.g., 2007 Q1), which is a restrictive specification that tests whether our inferences hold within firm; and (3) Fama–Macbeth estimation with no fixed effects, where we estimate our empirical models by year and compute standard errors based on the standard deviation of annual coefficient estimates.

4.1. Primary analysis

Table 2 reports the results from the estimation of Eq. (1) where the information release considered is an analyst forecast. In our full sample, average bid-ask spread two days prior to an analyst forecast announcement is 87.8 basis points (when *Price*, *LnSize*, *Volatility*, *Turnover*, *Depth*, *Volume* and *AbsCAR* are at their mean values). Focusing on column (3) for discussion, the *Info0^{AF}* coefficient of -0.842 (t -statistic -5.96) therefore indicates an approximate 1% reduction in bid-ask spread on the day of an analyst forecast announcement relative to two days prior to the announcement (i.e., $-0.842/87.8$). Consistent with our expectation that the announcement-period effects extend into the day after the recorded announcement date, the *Info1^{AF}* coefficient of -1.296 (t -statistic -7.75) indicates that bid-ask spread decreases further by -0.454 (i.e., from -0.842 to -1.296) on the day after the forecast announcement to a total reduction (relative to two days prior to announcement) of approximately 1.5%. These findings are consistent with our prediction that analyst forecasts, on average, immediately decrease information asymmetry at announcement. We note that inferences are consistent across all alternative fixed effect and standard error structures considered, as reported in columns (1), (2), and (4).¹¹ We also note that the insignificant coefficient on *PreInfo1^{AF}* suggests that on average there is no pre-announcement leakage of the content of analyst forecasts.

¹⁰ Subsequently reported inferences are unaltered if we delete observations within plus or minus five days of an earnings announcement or management forecast.

¹¹ For the remainder of our analyses, for brevity we tabulate only the firm-fixed-effects specification, as inferences are identical using the alternative year-fixed-effects and Fama–Macbeth specifications.

Table 1
Descriptive statistics and correlations.

Panel A: Descriptive statistics									
Variable	N	Mean	Std Dev	P25	P50	P75			
<i>BASpread</i>	2,158,116	86.827	129.420	20.312	47.158	100.445			
<i>Depth</i>	2,158,116	43.679	87.720	8.334	18.357	38.271			
<i>PriceImpact</i>	551,649	10.618	8.047	5.632	8.654	13.176			
<i>Price</i>	2,158,116	30.244	24.638	13.910	24.910	39.938			
<i>LnSize</i>	2,158,116	7.261	1.723	6.060	7.178	8.370			
<i>Volatility</i>	2,158,116	0.029	0.017	0.017	0.024	0.035			
<i>Turnover</i>	2,158,116	9.472	9.856	3.264	6.377	12.020			
<i>Volume</i>	2,158,116	1.421	3.508	0.103	0.349	1.128			
<i>AbsCAR</i>	2,158,116	0.029	0.034	0.008	0.018	0.037			
Panel B: Correlations									
	<i>BASpread</i>	<i>Depth</i>	<i>PriceImpact</i>	<i>Price</i>	<i>LnSize</i>	<i>Volatility</i>	<i>Turnover</i>	<i>Volume</i>	<i>AbsCAR</i>
<i>BASpread</i>		0.026	0.000	−0.306	−0.517	0.256	−0.172	−0.178	0.204
<i>Depth</i>	0.170		−0.188	−0.105	0.157	−0.069	−0.062	0.270	−0.038
<i>PriceImpact</i>	−0.045	−0.258		0.450	0.157	0.157	0.205	0.132	0.086
<i>Price</i>	−0.476	−0.132	0.568		0.590	−0.300	0.002	0.038	−0.184
<i>LnSize</i>	−0.680	0.183	0.246	0.687		−0.367	0.022	0.437	−0.252
<i>Volatility</i>	0.311	−0.082	0.069	−0.469	−0.421		0.439	0.052	0.379
<i>Turnover</i>	−0.364	−0.190	0.180	−0.033	0.065	0.459		0.261	0.160
<i>Volume</i>	−0.651	0.163	0.190	0.211	0.659	0.028	0.518		
<i>AbsCAR</i>	0.223	−0.023	0.052	−0.242	−0.245	0.364	0.155	0.010	

Panel A of Table 1 presents descriptive statistics for continuous variables used in our analyses. Panel B of Table 1 presents correlations among key variables of interest, with Pearson (Spearman) correlations reported above (below) the diagonal. Variable definitions are presented in the Appendix A. Statistics are based on our full analyst forecast sample, except for *PriceImpact*, which is only available for a smaller subsample.

4.2. Earnings announcements and management forecasts

As previously discussed, extant literature documents an announcement-period *increase* in information asymmetry for earnings announcements and management earnings forecasts. In this section, we reproduce these prior inferences using our sample and methodology. This analysis makes explicit the differences between the information asymmetry effects of information releases that have a relatively high concentration of information that is new to all market participants (i.e., earnings announcements and management forecasts), and those effects of information releases that have a relatively high concentration of information that is new only to unsophisticated investors (i.e., analyst forecasts). Specifically, we analyze a subsample of firm-quarters that contain all three information release types. Although this materially reduces our sample size (primarily because of the management forecast requirement), this empirical strategy effectively allows us to use a firm as its own unit of comparison.

Table 3 reports results from estimating Eq. (1) using this subsample, where Columns (1), (2), and (3) analyze analyst forecasts, earnings announcements, and management forecasts, respectively. Column (1) again documents that analyst forecasts decrease announcement-period information asymmetry ($Info0^{AF}$ coefficient -0.858 ; t -statistic -6.65). Similar to our results using the full sample, an analyst forecast decreases bid-ask spread on the day of the announcement by approximately 1% relative to two days prior to the announcement. Because there are between two and three analyst forecasts in each quarter, the overall quarterly economic effect of analyst forecasts on a given firm's bid-ask spread is an approximate 2.5% decrease, on average.

In contrast, column (2) documents that earnings announcements *increase* announcement-period information asymmetry. The $Info0^{EA}$ coefficient of 4.482 (t -statistic 9.35) indicates an approximate 5% increase in bid-ask spread on the day of an earnings announcement relative to two days prior to the announcement (i.e., $4.482/87.8$). Column (3) similarly documents that management forecasts increase announcement-period information asymmetry ($Info0^{MF}$ coefficient 4.398; t -statistic 12.67), with an economic magnitude similar to earnings announcements. Taken together, these results suggest the magnitude of the decrease in information asymmetry on the day of an analyst forecast announcement is about one-half the magnitude of the information asymmetry increase on the announcement day of earnings announcements and management forecasts.¹² These results are consistent with prior literature (e.g., [Coller and Yohn, 1997](#)), and with our prediction above that earnings announcements and management forecasts increase information asymmetry at announcement, in contrast to the directionally opposite effect of analyst forecasts.¹³ Also, we note that the positive and significant coefficient estimates on

¹² This is a conservative comparison. If we repeat this comparison using the effects of both day zero and day one (the full announcement window we use in our empirical tests), the effects of analyst forecasts are comparable in magnitude to the effects of earnings announcements and management forecasts.

¹³ In untubulated analyses, we consider only management forecasts that were not bundled with earnings announcements (and vice versa) and obtain identical inferences.

Table 2

Primary results – analyst forecasts and information asymmetry.

Dep. Var: Column:	<i>BASpread</i> (1)	<i>BASpread</i> (2)	<i>BASpread</i> (3)	<i>BASpread</i> (4)
<i>Intercept</i>		338.159*** (21.60)		312.915*** (14.92)
<i>PreInfo1^{AF}</i>	0.075 (0.98)	0.071 (0.66)	0.100 (0.93)	0.057 (0.69)
<i>Info0^{AF}</i>	−0.966*** (−11.29)	−0.970*** (−6.67)	−0.842*** (−5.96)	−0.943*** (−6.88)
<i>Info1^{AF}</i>	−1.212*** (−13.41)	−1.274*** (−7.67)	−1.296*** (−7.75)	−1.202*** (−6.79)
<i>Price</i>	0.266*** (8.78)	0.396*** (6.19)	0.327*** (6.63)	0.201* (1.82)
<i>LnSize_{t−1}</i>	−38.434*** (−56.80)	−39.337*** (−21.31)	−54.095*** (−28.69)	−36.076*** (−13.88)
<i>Volatility_{t−1}</i>	1,109.762*** (24.49)	1,356.130*** (8.21)	916.105*** (5.41)	1,670.156*** (5.26)
<i>Turnover_{t−1}</i>	−2.917*** (−36.32)	−3.477*** (−24.35)	−2.220*** (−19.28)	−3.201*** (−14.22)
<i>Depth</i>	0.067*** (9.73)	0.136*** (12.95)	0.013 (1.55)	0.132** (2.23)
<i>Volume</i>	4.140*** (21.53)	2.983*** (11.47)	0.879*** (3.58)	5.204*** (7.93)
<i>AbsCAR</i>	183.728*** (22.53)	232.542*** (10.66)	184.757*** (10.63)	168.629*** (5.38)
Fixed effects	Year	No	Firm	No
Clustered SE	Firm	Firm, Quarter	Firm, Quarter	FM
<i>N</i>	2,158,116	2,158,116	2,158,116	107,906
<i>Adj. R²</i>	0.365	0.340	0.565	0.330

Table 2 presents results of OLS estimation of Eq. (1) using our full analyst forecast sample. *PreInfoN* is an indicator variable that equals one if trading day *d* is the *N*th trading day preceding the announcement day of an information release, and equals zero otherwise. *InfoN* is an indicator variable that equals one if trading day *d* is the information release announcement date (*N*=0) or the trading day subsequent to the announcement date (*N*=1) and equals zero otherwise. All variable definitions are presented in the Appendix A. For Fama-MacBeth regressions (indicated where Clustered SE=FM), coefficient estimates, *N*, and *Adj. R²* are averages across the yearly regressions, and *t*-statistics are based on the standard deviation of annual coefficient estimates. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels, respectively. Intercepts and fixed effects coefficients are included but not reported for fixed effects estimations.

PreInfo1^{IR} suggest that for earnings announcements and management forecasts, there is some pre-announcement leakage or anticipation of their information content.

One potential concern with the above analysis is that, because *BASpread* mechanically decreases as stock price increases, our results may be driven by price increases during the forecast announcement period if analyst forecasts exhibit positive bias, on average. A second related concern is that several prior studies (e.g., McNichols and O'Brien, 1997) show that analysts prefer to deliver good news and may censor negative opinions. Further, this censoring may not occur in the case of earnings announcements and management forecasts. To address these concerns, we re-estimate Eq. (1) separately for sample partitions based on whether the forecast is good or bad news relative to the outstanding mean consensus forecast. As reported in columns (4) and (5) of Table 3, our inferences hold in both good news and bad news forecast sample partitions.¹⁴ The fact that our results obtain regardless whether analysts report good or bad news forecasts makes it unlikely that our findings are driven by either forecast bias or censoring related to the nature of the news.

4.3. Longer-window information asymmetry effects

Our primary interest in this study is the effect of information releases on short-window announcement-period information asymmetry. However, in this section we analyze the effects of information releases on information asymmetry in a longer window around the announcement date by estimating the following modification to Eq. (1):

$$\begin{aligned}
 \text{BASpread}_{i,d} = & \beta_0 + \sum_{N=3}^{10} \delta_N \text{PreInfoN}_{i,d}^{\text{IR}} + \beta_1 \text{PreInfo1}_{i,d}^{\text{IR}} + \beta_2 \text{Info0}_{i,d}^{\text{IR}} + \beta_3 \text{Info1}_{i,d}^{\text{IR}} \\
 & + \sum_{N=2}^{10} \varphi_N \text{PostInfoN}_{i,d}^{\text{IR}} + \beta_4 \text{Price}_{i,d} + \beta_5 \text{LnSize}_{i,t-1} + \beta_6 \text{Volatility}_{i,t-1} \\
 & + \beta_7 \text{Turnover}_{i,t-1} + \beta_8 \text{Depth}_{i,d} + \beta_9 \text{Volume}_{i,d} + \beta_{10} \text{AbsCAR}_{i,t} + \varepsilon_{i,d}.
 \end{aligned} \quad (2)$$

¹⁴ In untabulated analysis, we repeat our tests using unscaled bid-ask spread, and our inferences are robust. Further, we confirm that the inferences with respect to both earnings announcements and management forecasts also are identical across good and bad news sample partitions.

Table 3

Analyst forecasts, management forecasts, earnings announcements, and information asymmetry.

Info (IR): Dep. Var: Sample: Column:	AF BASpread (1)	EA BASpread (2)	MF BASpread (3)	AF BASpread Good News (4)	AF BASpread Bad News (5)
<i>PreInfo1^{IR}</i>	−0.119 (−1.16)	1.228*** (4.50)	0.919*** (3.89)	−0.151 (−0.85)	−0.065 (−0.39)
<i>Info0^{IR}</i>	−0.858*** (−6.65)	4.482*** (9.35)	4.398*** (12.67)	−0.740*** (−3.86)	−0.899*** (−5.04)
<i>Info1^{IR}</i>	−1.230*** (−9.37)	2.638*** (5.06)	2.178*** (5.77)	−1.079*** (−4.66)	−1.239*** (−8.05)
<i>Price</i>	0.150*** (2.76)	0.094 (0.94)	0.122* (1.74)	0.177*** (3.44)	0.162** (2.38)
<i>LnSize_{t−1}</i>	−31.011*** (−18.90)	−40.869*** (−16.25)	−36.532*** (−16.76)	−26.904*** (−14.80)	−33.402*** (−18.97)
<i>Volatility_{t−1}</i>	739.996*** (5.49)	654.839*** (4.65)	846.663*** (7.13)	637.591*** (4.96)	782.084*** (5.42)
<i>Turnover_{t−1}</i>	−1.601*** (−10.88)	−1.658*** (−10.76)	−2.001*** (−13.10)	−1.426*** (−10.28)	−1.742*** (−9.96)
<i>Depth</i>	0.061 (1.55)	0.026 (0.48)	0.096 (1.62)	0.051 (1.37)	0.052 (1.26)
<i>Volume</i>	−0.722*** (−3.70)	−0.185 (−1.28)	−0.213** (−2.02)	−0.316** (−2.32)	−0.222** (−2.19)
<i>AbsCAR</i>	114.759*** (9.39)	29.341*** (3.35)	19.052*** (2.72)	88.180*** (5.66)	116.145*** (8.51)
Fixed effects	Firm	Firm	Firm	Firm	Firm
Clustered SE	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter
<i>N</i>	197,972	76,804	91,788	66,924	130,732
<i>Adj. R²</i>	0.599	0.607	0.580	0.606	0.612

Table 3 presents results of OLS estimation of Eq. (1) separately for analyst forecasts, management forecasts, and earnings announcements using a subsample of firm-quarters that contain all three information release types. The last two columns present separate estimations of Eq. (1) for good news vs. bad news analyst forecasts (relative to the outstanding mean consensus analyst forecast). *PreInfoN* is an indicator variable that equals one if trading day *d* is the *Nth* trading day preceding the announcement day of an information release, and equals zero otherwise. *InfoN* is an indicator variable that equals one if trading day *d* is the information release announcement date (*N*=0) or the trading day subsequent to the announcement date (*N*=1) and equals zero otherwise. All variable definitions are presented in the Appendix A. Robust t-statistics are reported in parentheses. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels, respectively. Intercepts and fixed effects coefficients are included but not reported for fixed effects estimations.

PostInfoN^{IR} (*PreInfoN^{IR}*) is an indicator variable that equals one if day *d* is trading day *N* after (before) the announcement day (*Info0^{IR}*) and equals zero otherwise, where again in this specification the trading day two days prior to the announcement day (*PreInfo2*) is the excluded day captured by the intercept, and all other *PostInfoN^{IR}* and *PreInfoN^{IR}* coefficients are estimated incremental to that day. All other variables are as previously defined.

Table 4 presents the results from estimating Eq. (2) using the subsample of firm-quarters which contain all three information release types, where columns (1), (2), and (3) report results for analyst forecasts, earnings announcements, and management forecasts, respectively. Focusing on column (1), the significantly negative coefficient estimates for the *PostInfoN^{AF}* variables provide evidence that after an analyst forecast announcement, information asymmetry remains lower than observed prior to the announcement, consistent with our expectation. The generally insignificant coefficients on the *PreInfoN* variables reveal that information asymmetry is relatively stationary in the nine days leading up to the release of an analyst forecast. This evidence that there is not a noteworthy spike in information asymmetry in advance of analyst forecast announcements helps alleviate concern that analyst forecasts are simply piggybacking on recent news events.

Focusing on earnings announcements and management forecasts in columns (2) and (3), respectively, the significantly negative coefficient estimates for the *PostInfoN* variables provide evidence that, as with analyst forecasts, after the initial announcement-period increase in information asymmetry there is a decrease in information asymmetry relative to the pre-announcement levels, consistent with prior literature (e.g., Coller and Yohn, 1997; Lee et al., 1993).

Fig. 1, which depicts these regression results graphically, clearly reveals the directionally opposite announcement-window information asymmetry effects of analyst forecasts relative to both management forecasts and earnings announcements. Again, whereas all three information releases decrease information asymmetry in the post-announcement window, analyst forecasts are distinct in that the decrease in information asymmetry begins at announcement.

4.4. Alternative measures of information asymmetry

Extant literature often uses quoted depth and price impact of trades as alternative measures of information asymmetry. Specifically, an increase (decrease) in quoted depth reflects a decrease (increase) in information asymmetry, and an increase

(decrease) in price impact reflects an increase (decrease) in information asymmetry. In this section, we repeat the above analysis with these alternative measures as our dependent variable.

In columns (4)–(6) of Table 4, we report results from a modification of Eq. (2) where we use *Depth* as the dependent variable (as defined previously), where in this specification we use *BASpread* as a control variable. As reported, our inferences are unchanged relative to columns (1)–(3). Specifically, at announcement, *Depth* increases for analyst forecasts and decreases for both earnings announcements and management forecasts. Fig. 2 depicts these results graphically.

In columns (7)–(9) of Table 4, we report results from a modification of Eq. (2) where we use *PricelImpact* as the dependent variable (as defined in the Appendix A).¹⁵ As reported, our inferences are unchanged relative to columns (1)–(3). Specifically, at announcement, *PricelImpact* decreases for analyst forecasts and increases for both earnings announcements and management forecasts. Fig. 3 depicts these results graphically.

Despite the fact that our inferences are consistent when using either quoted depth or price impact, prior literature suggests that bid-ask spread is a superior measure of information asymmetry, as the alternatives are much more likely to contain significant measurement error (e.g., Amihud, 2002; Bacidore et al., 2002; Heflin et al., 2005). Therefore, in subsequent analyses, for brevity we focus our attention to the use of *BASpread* as our dependent variable, with *Depth* included as a control. Moreover, for our remaining analyses we return to our full analyst forecast sample, with no additional comparisons made to earnings announcements or management forecasts.

4.5. Anticipation of information releases

One concern about the previous comparative analysis is that, relative to earnings announcements and management forecasts, analyst forecasts are more likely to be unanticipated, and this difference may contribute to our findings. Indeed, prior literature finds evidence of predictable behavior by market makers around anticipated events, such as earnings announcements (e.g., So and Wang, 2014). Intuitively, to address this concern we would like to show that information asymmetry also decreases at announcement of analyst forecasts that are anticipated. Extant literature documents that it is commonplace for analysts to issue forecasts in the week after an earnings announcement (e.g., Bagnoli et al., 2005). Therefore, our empirical strategy is to examine the information asymmetry effects of analyst forecasts that are issued in the week after an earnings announcement, as those are more likely to be anticipated.

Table 5 reports the results from estimating Eq. (1) separately for analyst forecasts issued in the week following an earnings announcement (i.e., “Anticipated”) and issued at other times (i.e., “Unanticipated”), where we continue to exclude analyst forecasts that are issued within two days after an earnings announcement to avoid contamination. As reported, the decrease in information asymmetry at announcement is significant for both categories, which suggests that our results are not attributable to differences in anticipation of the alternative information release types.

4.6. Addressing endogeneity concerns using non-clustered forecasts

Another potential concern is that rather than analyst forecasts causing a decrease in information asymmetry, our findings may reflect an outcome of analysts issuing forecasts when information asymmetry is already decreasing (i.e., reverse causality). For example, if analysts piggyback and issue forecasts immediately after a significant firm news event (e.g., Altinkiliç and Hansen, 2009), we may be simply picking up the natural resolution of information asymmetry from that news event, rather than a causal effect from analyst forecasts.¹⁶

As previously discussed, the evidence in Table 4 and Fig. 1 suggests that this alternative explanation is unlikely, because there is no evidence of a material spike in information asymmetry immediately prior to analyst forecasts. Further, recent evidence suggests that analysts do not simply piggyback on news events (e.g., Li et al., 2015). However, to further mitigate this concern, we consider an additional test based on the following logic—if one of the reverse causality stories holds, we would expect clustering of forecasts as multiple analysts following a firm react to the same news stimulus. Accordingly, we repeat our analysis using only analyst forecasts where there is no other analyst forecast for the same firm within a five-day surrounding window, after limiting the sample to firm-quarters with multiple analysts following the firm (we explore both greater than three and greater than ten analysts). The resulting sub-sample then consists of analyst forecasts that are issued in relative isolation with respect to other analyst forecasts for the same firm-quarter.

Columns (1) and (2) of Table 6 present results from this analysis for firm-quarters with at least three and ten forecasting analysts, respectively. To summarize, results are consistent with our primary analysis, i.e., there is a significantly negative coefficient for both *Info0^{AF}* and *Info1^{AF}*. Because our results obtain after eliminating clustered forecasts, the effects we document are unlikely to be driven by preceding information events or analysts' reaction to decreasing information asymmetry. We also note that the diminished coefficient magnitudes in this analysis are expected, as the coefficient

¹⁵ The price impact specification is estimated using a subsample for which we have the *PricelImpact* measure available (spanning the years 2001–2007). We have no reason to believe that results from this specification would differ using our full sample period. Furthermore, we re-estimated Eq. (2) using both *BASpread* and *Depth* as dependent variables while limiting those analyses to the 2001–2007 sample period, and inferences are identical to our previously reported analyses using the full sample period.

¹⁶ Because of our sample construction, we remove this possibility with respect to earnings announcements and management forecasts. Therefore, the potential concern would stem from other types of news events that are less easily observable.

Table 4

Information releases and longer-window information asymmetry.

Info (IR):	AF	EA	MF	AF	EA	MF	AF	EA	MF
Dep. Var:	BASpread			Depth			PricelImpact		
Column:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>PreInfo10^{IR}</i>	−0.451*	−0.138	−0.173	−0.062	0.359***	0.113	−0.219***	−0.218*	−0.067
	(−1.85)	(−0.28)	(−0.48)	(−0.73)	(2.95)	(1.05)	(−3.59)	(−1.93)	(−0.50)
<i>PreInfo9^{IR}</i>	−0.356	0.163	−0.054	−0.121	0.278***	0.025	−0.149***	0.067	−0.141
	(−1.63)	(0.35)	(−0.14)	(−1.59)	(2.60)	(0.28)	(−2.23)	(0.33)	(−1.19)
<i>PreInfo8^{IR}</i>	−0.297	0.379	−0.317	−0.147*	0.218**	0.065	−0.164***	−0.040	−0.236**
	(−1.47)	(0.95)	(−0.90)	(−1.92)	(2.04)	(0.69)	(−3.18)	(−0.28)	(−2.50)
<i>PreInfo7^{IR}</i>	−0.154	0.052	−0.545**	−0.067	0.284***	0.089	−0.148**	−0.073	−0.246*
	(−0.76)	(0.18)	(−1.96)	(−0.90)	(2.85)	(1.15)	(−2.55)	(−0.40)	(−1.86)
<i>PreInfo6^{IR}</i>	−0.189	0.207	−0.345	−0.086	0.130	−0.002	−0.193***	−0.074	−0.257***
	(−1.02)	(0.72)	(−1.05)	(−1.21)	(1.41)	(−0.03)	(−3.30)	(−0.66)	(−3.74)
<i>PreInfo5^{IR}</i>	−0.283*	−0.050	−0.194	−0.025	0.065	0.012	−0.142***	0.112	−0.053
	(−1.83)	(−0.18)	(−0.79)	(−0.38)	(0.78)	(0.17)	(−2.95)	(0.80)	(−0.66)
<i>PreInfo4^{IR}</i>	−0.314**	0.515*	0.043	−0.022	0.103	−0.026	−0.156***	0.183	−0.066
	(−2.26)	(1.89)	(0.18)	(−0.40)	(1.11)	(−0.32)	(−3.85)	(1.32)	(−0.90)
<i>PreInfo3^{IR}</i>	−0.230**	0.652***	0.228	0.025	0.063	−0.035	0.053	0.152	−0.024
	(−2.24)	(3.33)	(1.33)	(0.51)	(0.79)	(−0.56)	(0.91)	(1.44)	(−0.37)
<i>PreInfo1^{IR}</i>	−0.116	1.239***	0.942***	0.091	−0.412***	−0.329***	0.025	0.715***	0.608***
	(−1.14)	(4.60)	(4.05)	(1.56)	(−4.52)	(−3.88)	(0.39)	(10.71)	(10.53)
<i>Info0^{IR}</i>	−0.853***	4.617***	4.734***	0.293***	−1.232***	−0.539*	−0.134**	4.303***	4.373***
	(−6.66)	(9.23)	(12.86)	(4.79)	(−4.29)	(−1.67)	(−2.53)	(20.94)	(24.11)
<i>Info1^{IR}</i>	−1.215***	2.869***	2.672***	0.250***	−0.731*	0.161	−0.390***	4.371***	3.978***
	(−9.34)	(5.12)	(6.27)	(3.40)	(−1.85)	(0.39)	(−8.55)	(14.62)	(15.90)
<i>PostInfo2^{IR}</i>	−0.806***	−1.804***	−1.710***	0.140**	1.193***	1.433***	−0.438***	0.686***	0.556***
	(−4.94)	(−4.26)	(−5.28)	(2.01)	(7.79)	(9.00)	(−8.65)	(6.90)	(6.03)
<i>PostInfo3^{IR}</i>	−0.524***	−1.583***	−1.881***	−0.027	0.989***	1.199***	−0.296***	0.132	−0.143
	(−3.05)	(−3.21)	(−5.92)	(−0.42)	(5.96)	(9.82)	(−5.54)	(1.56)	(−1.32)
<i>PostInfo4^{IR}</i>	−0.425**	−1.768***	−2.127***	0.006	1.084***	1.261***	−0.370***	−0.239**	−0.259**
	(−2.13)	(−3.60)	(−6.10)	(0.08)	(6.51)	(8.81)	(−5.72)	(−2.16)	(−2.49)
<i>PostInfo5^{IR}</i>	−0.646***	−1.953***	−1.776***	0.083	0.924***	1.075***	−0.292***	−0.196**	−0.284**
	(−3.09)	(−3.42)	(−4.29)	(1.00)	(5.32)	(8.60)	(−4.02)	(−2.05)	(−2.22)
<i>PostInfo6^{IR}</i>	−0.545***	−1.316**	−1.237***	0.017	0.797***	0.828***	−0.317***	−0.199	−0.315**
	(−2.59)	(−2.39)	(−2.90)	(0.22)	(4.79)	(7.12)	(−4.84)	(−1.57)	(−2.51)
<i>PostInfo7^{IR}</i>	−0.553**	−1.656***	−1.663***	0.015	0.838***	0.895***	−0.365***	−0.046	−0.455***
	(−2.48)	(−2.81)	(−4.07)	(0.17)	(5.39)	(8.17)	(−4.94)	(−0.31)	(−3.17)
<i>PostInfo8^{IR}</i>	−0.545**	−1.707***	−1.424***	0.035	0.708***	0.756***	−0.280***	−0.331**	−0.422***
	(−2.35)	(−2.97)	(−2.87)	(0.47)	(3.92)	(5.86)	(−3.35)	(−2.00)	(−2.85)
<i>PostInfo9^{IR}</i>	−0.514**	−1.567***	−1.513***	−0.023	0.689***	0.762***	−0.376***	−0.414**	−0.395***
	(−1.97)	(−2.60)	(−3.57)	(−0.26)	(3.72)	(6.50)	(−4.66)	(−2.48)	(−2.66)
<i>PostInfo10^{IR}</i>	−0.537**	−1.713***	−1.530***	0.022	0.635***	0.713***	−0.429***	−0.298*	−0.312*
	(−2.18)	(−2.89)	(−3.47)	(0.25)	(3.49)	(5.71)	(−4.93)	(−1.72)	(−1.88)
<i>Price</i>	0.151***	0.106	0.140**	−0.280***	−0.233***	−0.266***	0.310***	0.316***	0.331***
	(2.73)	(1.18)	(2.21)	(−5.54)	(−5.93)	(−7.02)	(13.71)	(11.17)	(11.27)
<i>LnSize_{t−1}</i>	−31.411***	−40.232***	−36.243***	−4.520***	−4.589***	−3.052***	−0.602	−0.541	−0.738*
	(−19.64)	(−17.33)	(−17.51)	(−3.62)	(−5.24)	(−3.44)	(−1.50)	(−1.54)	(−1.83)
<i>Volatility_{t−1}</i>	754.848***	650.975***	842.499***	−97.280***	−86.659***	−50.671	224.841***	188.882***	212.690***
	(5.52)	(4.71)	(7.13)	(−2.75)	(−3.06)	(−1.59)	(7.85)	(7.56)	(7.41)
<i>Turnover_{t−1}</i>	−1.629***	−1.578***	−1.940***	−0.252***	−0.128**	−0.183***	−0.039	−0.037	−0.042
	(−11.09)	(−10.58)	(−13.20)	(−3.32)	(−2.33)	(−2.97)	(−1.59)	(−1.54)	(−1.46)
<i>Volume</i>	−0.545***	−0.304	−0.454***	1.357***	1.269***	0.993***	0.725***	0.552***	0.614***
	(−3.01)	(−1.56)	(−2.93)	(3.89)	(5.82)	(4.28)	(8.44)	(7.15)	(8.09)
<i>AbsCAR</i>	88.941***	16.833**	13.934**	−6.642	−2.865	7.931***	17.495***	10.047***	7.788***
	(8.24)	(2.15)	(2.08)	(−0.98)	(−1.44)	(2.98)	(9.16)	(8.87)	(9.95)
<i>Depth</i>	0.054	−0.008	0.046						
	(1.36)	(−0.18)	(0.86)						
<i>BASpread</i>				0.011	−0.001	0.005			
				(1.27)	(−0.18)	(0.83)			
Fixed effects	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm
Clustered SE	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter	Firm,Quarter
<i>N</i>	1,039,353	403,221	481,887	1,039,353	403,221	481,887	551,649	227,934	284,886
<i>Adj. R²</i>	0.597	0.607	0.578	0.605	0.599	0.605	0.484	0.399	0.463

Table 4 presents results of OLS estimation of Eq. (2) separately for analyst forecasts, management forecasts, and earnings announcements using a subsample of firm-quarters containing all three information release types. *PreInfoN* is an indicator variable that equals one if trading day *d* is the *N*th trading day preceding the announcement day of an information release, and equals zero otherwise. *InfoN* is an indicator variable that equals one if trading day *d* is the information release announcement date (*N*=0) or the trading day subsequent to the announcement date (*N*=1) and equals zero otherwise. *PostInfoN* is an indicator variable that equals one if trading day *d* is the *N*th trading day after the announcement day of an information release, and equals zero otherwise. All variable definitions are presented in the Appendix A. Robust t-statistics are reported in parentheses. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels. Intercepts and fixed effects are included but not reported.

estimates capture the average effect on information asymmetry of *one* analyst forecast. In this subsample of firm-quarters where there are a large number of analyst forecasts, we naturally expect each individual forecast to have a smaller average effect.

4.7. Forecast information content

To buttress our main findings regarding the effect of analyst forecasts on information asymmetry, we next examine cross-sectional variation based on several alternative measures of forecast information content. Specifically, we test whether the effect of an analyst forecast on information asymmetry is larger when there is greater information content in the forecast by estimating the following model:

$$\begin{aligned} BASpread_{i,d} = & \beta_0 + \beta_1 Info01_{i,d}^{AF} + \beta_2 Content_{i,r} + \beta_3 Content_{i,r} * Info01_{i,d}^{AF} + \beta_4 Price_{i,d} \\ & + \beta_5 LnSize_{i,t-1} + \beta_6 Volatility_{i,t-1} + \beta_7 Turnover_{i,t-1} + \beta_8 Depth_{i,d} \\ & + \beta_9 Volume_{i,d} + \beta_{10} AbsCAR_{i,r} + \varepsilon_{i,d}, \end{aligned} \quad (3)$$

where *Content* takes on three alternative forms, as described below. Our previous results imply that the announcement window for analyst forecasts includes both the measured announcement day and the subsequent day. Therefore, for this test we define $Info01^{AF}$ as an indicator variable that equals one if $Info0^{AF} = 1$ or $Info1^{AF} = 1$. $Info01^{AF}$ captures the average change in information asymmetry across the announcement day and the subsequent day (relative to the average level of information asymmetry for the two trading days immediately preceding the announcement day). For each *Content* measure, we predict $\beta_3 < 0$, which would indicate that an analyst forecast decreases announcement-period information asymmetry to a greater extent with there is relatively high information content in the forecast.

To fix intuition, take the information set of sophisticated investors as given with respect to a firm, which sophisticated investors obtain as a result of costly effort. The more effort an analyst exerts in processing information contained within a given forecast for that firm, the better that forecast will approximate the information set of sophisticated investors, which will result in greater reduction in information asymmetry. Of course, if the analyst exerts *more* effort than the sophisticated investors, there is an increased likelihood that the sophisticated investors might themselves learn new information from the analyst forecast, which will bring the asymmetry-increasing force into play. However, we expect the increase in the asymmetry-decreasing force to be greater still, leading to a larger net decrease in information asymmetry.

First, we consider an effort-based measure of information content. We follow Barth et al. (2001) and consider the number of firms covered by an analyst to be an inverse proxy for analyst effort in covering a particular firm, i.e., the more firms an analyst covers, the less effort she can exert on any single firm-forecast. Accordingly, we define *Effort* as an indicator variable that equals one if the number of distinct firms covered by the forecasting analyst during the quarter in which the forecast is provided is below the sample median (i.e., reflecting relatively high effort in each forecast provided by that analyst), and equals zero otherwise. Second, we measure forecast information content as the individual forecast's deviation from the prevailing consensus forecast. We define *Deviation* as an indicator variable that equals one if the absolute percent

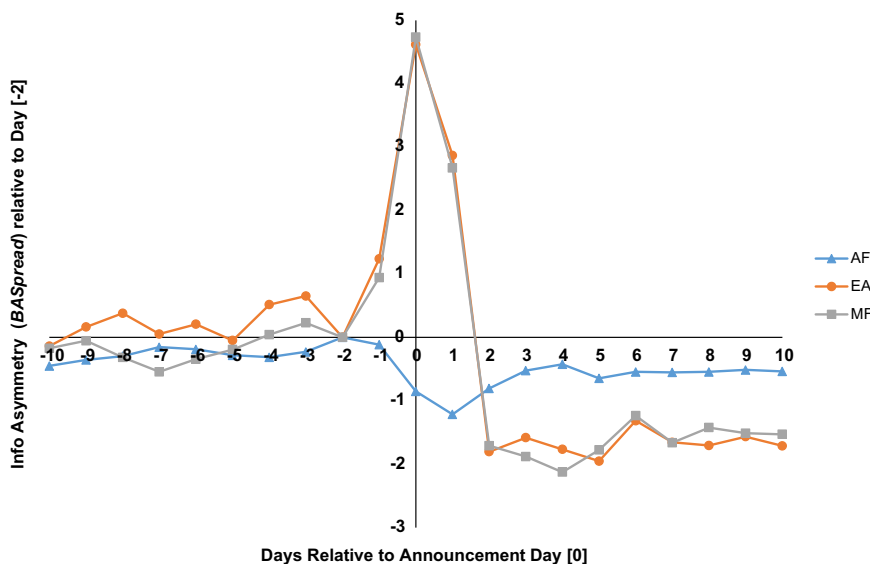


Fig. 1. Information asymmetry (bid-ask spread) around information releases relative to normalized baseline two days prior to the information release. Fig. 1 plots the $PreInfo10^{IR}$ through $PostInfo10^{IR}$ coefficient estimates from the Table 4 (columns 1–3) regression estimation for analyst forecasts, management forecasts, and earnings announcements. The horizontal axis reflects the timing of the variable construction (e.g., day 4 corresponds to the $PostInfo4^{IR}$ coefficient estimates).

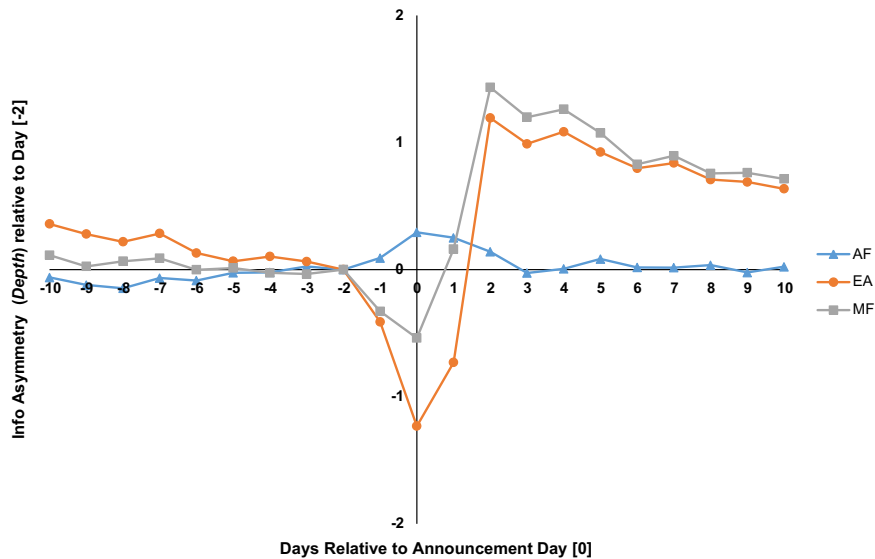


Fig. 2. Information asymmetry (depth) around information releases relative to normalized baseline two days prior to the information release. Fig. 2 plots the $PreInfo10^{IR}$ through $PostInfo10^{IR}$ coefficient estimates from the Table 4 (columns 4–6) regression estimation for analyst forecasts, management forecasts, and earnings announcements. The horizontal axis reflects the timing of the variable construction (e.g., day 4 corresponds to the $PostInfo4^{IR}$ coefficient estimates).

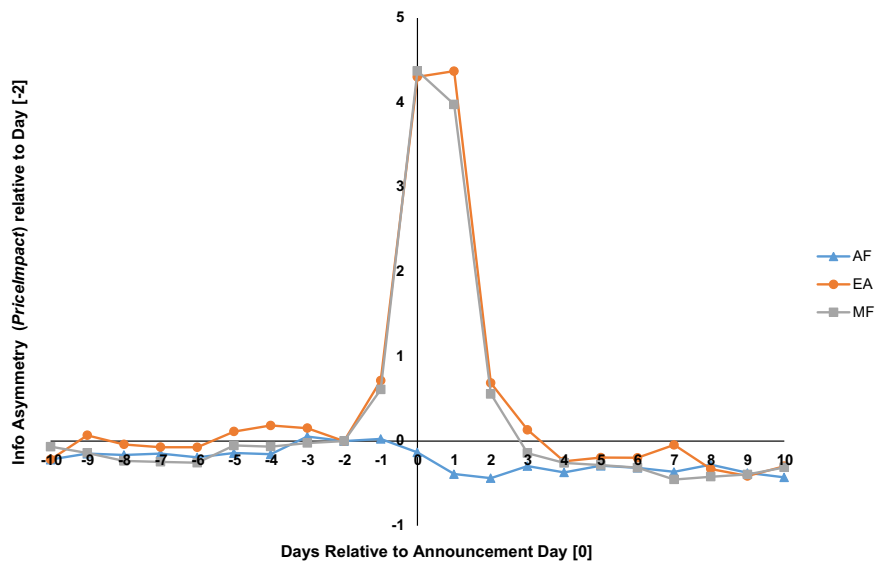


Fig. 3. Information asymmetry (price impact) around information releases relative to normalized baseline two days prior to the information release. Fig. 3 plots the $PreInfo10^{IR}$ through $PostInfo10^{IR}$ coefficient estimates from the Table 4 (columns 7–9) regression estimation for analyst forecasts, management forecasts, and earnings announcements. The horizontal axis reflects the timing of the variable construction (e.g., day 4 corresponds to the $PostInfo4^{IR}$ coefficient estimates).

deviation of an analyst's forecast from the prevailing consensus is above the sample median, and equals zero otherwise. Finally, we measure forecast information content as the magnitude of the forecast revision relative to the same analyst's previous forecast. Specifically, we define *Revision* as an indicator variable that equals one if the absolute percent revision in an analyst's forecast from that same analyst's most recent prior forecast for the same firm is above the sample median, and equals zero otherwise.

Table 7 presents results, which are consistent with our prediction using all three alternative measures of information content. That is, analyst forecasts reduce information asymmetry when information content is relatively low (i.e., $\beta_1 < 0$), and do so to a greater extent when information content is relatively high (i.e., $\beta_3 < 0$). Using the column (3) *Revision* results for discussion, the $Info01$ coefficient estimate is -0.929 (t -statistic -8.71), and the interaction coefficient is -0.349 (t -statistic -2.10). Results using *Effort* (column 1) and *Deviation* (column 2) are similar.

Table 5

Anticipated vs. unanticipated analyst forecasts.

Dep. Var.: Sample: Column:	<i>BASpread</i> Anticipated (1)	<i>BASpread</i> Unanticipated (2)
<i>PreInfo1^{AF}</i>	−0.925** (−2.50)	0.228** (2.38)
<i>Info0^{AF}</i>	−3.927*** (−7.73)	−0.453*** (−3.57)
<i>Info1^{AF}</i>	−4.433*** (−7.39)	−0.902*** (−5.83)
<i>Price</i>	0.387*** (5.75)	0.315*** (6.63)
<i>LnSize_{t-1}</i>	−58.310*** (−24.22)	−53.079*** (−28.80)
<i>Volatility_{t-1}</i>	1,315.412*** (8.66)	851.798*** (4.84)
<i>Turnover_{t-1}</i>	−2.988*** (−20.59)	−2.120*** (−18.26)
<i>Depth</i>	0.021* (1.77)	0.013 (1.53)
<i>Volume</i>	0.889*** (3.58)	0.853*** (3.45)
<i>AbsCAR</i>	188.717*** (7.24)	181.459*** (10.39)
Fixed effects	Firm	Firm
Clustered SE	Firm,Quarter	Firm,Quarter
<i>N</i>	240,672	1,917,444
<i>Adj. R²</i>	0.601	0.570

Table 5 presents results of OLS estimation of Eq. (1) separately for analyst forecasts issued within one week after an earnings announcement (“Anticipated”), and those issued at other times (“Unanticipated”), using our full analyst forecast sample. *PreInfoN* is an indicator variable that equals one if trading day *d* is the *N*th trading day preceding the announcement day of an information release, and equals zero otherwise. *InfoN* is an indicator variable that equals one if trading day *d* is the information release announcement date (*N*=0) or the trading day subsequent to the announcement date (*N*=1) and equals zero otherwise. All variable definitions are presented in the Appendix A. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels, respectively. Intercepts and fixed effects coefficients are included but not reported.

4.8. Differential forecast timing

Next, using an alternative construct that relates to forecast information content, we examine whether our results are sensitive to the timing of forecasts among analysts within a quarter. In a set of forecasts made for the same firm by different analysts in the same general time window, we expect that the earliest forecasts will reduce information asymmetry to a greater extent than later forecasts, as the earliest forecasts likely have the greatest information content (Cooper et al., 2001). Specifically, we use a subsample of the initial forecasts issued by each analyst for a given firm in each quarter *t* (where we restrict the subsample to forecasts issued after the quarter *t*−1 earnings announcement), and retain only the earliest and latest forecasts (i.e., each firm-quarter will then yield two forecast observations, made by different analysts). We then augment Eq. (1) by adding an indicator variable, *Last^{AF}*, and its interactions with our *InfoN^{AF}* variables, where *Last^{AF}* equals one (zero) for the later (earlier) of the two forecasts for each firm-quarter. In this resulting specification, the coefficients on *Info0^{AF}* and *Info1^{AF}* estimate the effect of the earliest forecasts on information asymmetry (which we expect to be negative, consistent with our preceding analyses), and their interactions with *Last^{AF}* estimate the incremental effect of the latest forecasts on information asymmetry. Because we expect the effect of later forecasts to be mitigated relative to earlier forecasts, we predict these interaction coefficients to be positive.

Table 8 reports results, which are consistent with our predictions. The *Info1^{AF}* coefficient of −1.791 (t-statistic −6.83) indicates that the earliest forecasts significantly decrease information asymmetry during the announcement window. Also, consistent with intuition, this effect from the earliest forecasts is of greater magnitude than the average effect across all forecasts documented in Table 2. Further, the *Info1^{AF}*Last^{AF}* interaction coefficient of 0.812 (t-statistic 3.25) is significantly positive, which indicates that the information-asymmetry-reducing effect of the latest forecasts (relative to the earliest forecasts) is mitigated. Further consistent with intuition, the total effect of the latest forecast (−1.791+0.812=−0.979) is of smaller magnitude than the average effect across all forecasts documented in Table 2.

5. Conclusion

Theoretical arguments in extant disclosure literature posit the existence of two countervailing forces on short-window information asymmetry between sophisticated and unsophisticated investors at the announcement of any given public

Table 6
Addressing endogeneity concerns using non-clustered forecasts.

Analyst Coverage: Dep. Var: Column:	> 3 <i>BASpread</i> (1)	> 10 <i>BASpread</i> (2)
<i>PreInfo1^{AF}</i>	0.078 (1.12)	0.053 (1.04)
<i>Info0^{AF}</i>	−0.668*** (−6.82)	−0.319*** (−4.59)
<i>Info1^{AF}</i>	−0.955*** (−8.14)	−0.474*** (−4.68)
<i>Price</i>	0.168*** (3.90)	0.017 (0.55)
<i>LnSize_{t−1}</i>	−40.614*** (−29.62)	−21.613*** (−17.57)
<i>Volatility_{t−1}</i>	708.440*** (4.40)	545.653*** (3.96)
<i>Turnover_{t−1}</i>	−1.823*** (−16.48)	−1.361*** (−11.89)
<i>Depth</i>	0.022*** (2.71)	0.031*** (4.06)
<i>Volume</i>	0.111 (0.66)	−0.455*** (−5.37)
<i>AbsCAR</i>	156.085*** (9.83)	121.170*** (8.47)
Fixed effects	Firm	Firm
Clustered SE	Firm,Quarter	Firm,Quarter
<i>N</i>	1,846,668	858,228
<i>Adj. R²</i>	0.502	0.391

Table 6 presents results of OLS estimation of Eq. (1) for a set of analyst forecasts where there are no other analyst forecasts issued for the same firm within a five-day surrounding window. *PreInfoN* is an indicator variable that equals one if trading day *d* is the *N*th trading day preceding the announcement day of an information release, and equals zero otherwise. *InfoN* is an indicator variable that equals one if trading day *d* is the information release announcement date (*N*=0) or the trading day subsequent to the announcement date (*N*=1) and equals zero otherwise. All variable definitions are presented in the Appendix A. Columns (1) and (2) use subsamples of firm-quarters with greater than three and ten forecasting analysts, respectively. Robust *t*-statistics are reported in parentheses. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels. Intercepts and fixed effects coefficients are included but not reported.

Table 7
Analyst forecast information content.

Dep. Var: Content: Column:	<i>BASpread</i> <i>Effort</i> (1)	<i>BASpread</i> <i>Deviation</i> (2)	<i>BASpread</i> <i>Revision</i> (3)
<i>Info01^{AF}</i>	−0.922*** (−6.25)	−0.898*** (−9.43)	−0.929*** (−8.71)
<i>Content</i>	0.672 (1.00)	−0.519 (−1.03)	−0.297 (−0.59)
<i>Content*Info01^{AF}</i>	−0.335** (−2.04)	−0.441*** (−2.68)	−0.349** (−2.10)
<i>Price</i>	0.327*** (6.65)	0.325*** (6.64)	0.345*** (7.52)
<i>LnSize_{t−1}</i>	−54.111*** (−28.65)	−54.153*** (−28.78)	−53.657*** (−27.33)
<i>Volatility_{t−1}</i>	916.063*** (5.42)	917.400*** (5.43)	812.431*** (4.79)
<i>Turnover_{t−1}</i>	−2.220*** (−19.30)	−2.219*** (−19.25)	−2.181*** (−16.68)
<i>Depth</i>	0.013 (1.56)	0.013 (1.55)	0.003*** (2.76)
<i>Volume</i>	0.878*** (3.57)	0.885*** (3.62)	0.705*** (3.45)
<i>AbsCAR</i>	184.653*** (10.61)	184.952*** (10.65)	160.671*** (9.33)
Fixed Effects	Firm	Firm	Firm
Clustered SE	Firm,Quarter	Firm,Quarter	Firm,Quarter
<i>N</i>	2,158,116	2,158,116	1,353,604
<i>Adj. R²</i>	0.565	0.565	0.602

Table 7 presents results of OLS estimation of Eq. (3) using our full analyst forecast sample. *Info01* is an indicator variable that equals one if trading day *d* is the information release announcement date or the trading day subsequent to the announcement date, and equals zero otherwise. *Content* is a variable that is increasing in analyst forecast information content, and takes three alternative forms—*Effort*, *Deviation*, and *Revision*. All variable definitions are presented in the Appendix A. Robust *t*-statistics are reported in parentheses. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels. Intercepts and fixed effects coefficients are included but not reported.

Table 8
Analyst forecast timing.

Dep. Var: Column:	BASpread (1)
<i>Last</i> ^{AF}	–0.459 (–0.65)
<i>PreInfo1</i> ^{AF}	–0.277* (–1.84)
<i>PreInfo1</i> ^{AF} * <i>Last</i> ^{AF}	0.425** (2.19)
<i>Info0</i> ^{AF}	–1.528*** (–7.08)
<i>Info0</i> ^{AF} * <i>Last</i> ^{AF}	0.958*** (4.36)
<i>Info1</i> ^{AF}	–1.791*** (–6.83)
<i>Info1</i> ^{AF} * <i>Last</i> ^{AF}	0.812*** (3.25)
<i>Price</i>	0.243*** (4.78)
<i>LnSize</i> _{<i>t</i>–1}	–46.754*** (–27.76)
<i>Volatility</i> _{<i>t</i>–1}	796.492*** (4.84)
<i>Turnover</i> _{<i>t</i>–1}	–2.031*** (–18.41)
<i>Depth</i>	0.022** (2.36)
<i>Volume</i>	0.404** (1.97)
<i>AbsCAR</i>	157.090*** (10.07)
Fixed effects	Firm
Clustered SE	Firm,Quarter
<i>N</i>	1,000,224
Adj. <i>R</i> ²	0.547

Table 8 presents results of OLS estimation of a modified version of Eq. (1) using a subsample of the earliest and latest initial forecasts issued by different analyst for firm *i* during quarter *t*. *PreInfoN* is an indicator variable that equals one if trading day *d* is the *N*th trading day preceding the announcement day of an information release, and equals zero otherwise. *InfoN* is an indicator variable that equals one if trading day *d* is the information release announcement date (*N*=0) or the trading day subsequent to the announcement date (*N*=1) and equals zero otherwise. *Last* is an indicator variable that equals one for the latest forecast issued for firm *i*, and equals zero for the earliest forecast issued for firm *i*. All variable definitions are presented in the Appendix A. Robust t-statistics are reported in parentheses. *, **, and *** indicate significance (two-sided) at the 10%, 5%, and 1% levels. Intercepts and fixed effects coefficients are included but not reported.

information release. An information release can increase information asymmetry if it contains information that is new to sophisticated investors, and can decrease information asymmetry if it contains information new to unsophisticated investors but already known by sophisticated investors. Although most information releases likely contain both categories of information to differing relative degrees, extant empirical literature has focused attention on studying the information asymmetry effects of information release types that have a relatively high concentration of information that is new to both sophisticated and unsophisticated investors (e.g., earnings announcements, management earnings forecasts).

As expected, this literature documents that both earnings announcements and management forecasts increase information asymmetry at announcement. Thus, there is no empirical evidence that the information asymmetry-decreasing force plays a significant role in announcement-period information dynamics. We use the setting of analyst forecasts to fill this gap. Compared to earnings announcements and management forecasts, extant literature suggests that analyst forecasts contain relatively more information that is new only to unsophisticated investors. Indeed, we document that analyst forecasts decrease information asymmetry at announcement. We further document that all three types of information releases (i.e., analyst forecasts, management forecasts and earnings announcements) decrease information asymmetry over the ten days subsequent to the announcement. Accordingly, as expected, the directionally opposite effects on information asymmetry between analyst forecasts and the other two information releases are temporary, and exist only during the short-window announcement period.

We do not intend or claim to provide evidence that analysts completely level the playing field between sophisticated and unsophisticated investors. Our only logical premise is that analyst forecasts contain a relatively high concentration of information that is new to unsophisticated investors (and relatively less information that is new to sophisticated investors).

That single premise, which is supported by the weight of extant literature, is sufficient to yield our predictions and explain our findings.

Our evidence yields insights into investor behavior and extant theories concerning information releases and information asymmetry. In particular, our evidence shows that different types of information releases lead to predictably opposite effects on information asymmetry at announcement, which provides some empirical validation to the theory that an information release can decrease announcement-period information asymmetry.

Appendix A. Variable definitions

$AbsCAR_{i,r}$	The absolute value of the cumulative abnormal return over the two-day announcement period for firm i 's information release r , where we calculate abnormal return as firm i 's return less the value-weighted market return.
$BASpread_{i,d}$	Firm i 's bid-ask spread in basis points on trading day d , measured as the mean daily bid minus ask, scaled by the midpoint. We use the consolidated quotes files from the TAQ database. The mean bid-ask spreads are calculated during regular trading hours, from 9:30 a.m. to 4:00 p.m..
$Depth_{i,d}$	Firm i 's quoted depth on trading day d , measured as the mean bid size plus offer size. Bid size and offer size represent 100 share lots. We use the consolidated quotes files from the TAQ database. The mean quoted depth is calculated during regular trading hours, from 9:30 a.m. to 4:00 p.m.
$Deviation_r$	An indicator variable that equals one if the absolute percent deviation of analyst forecast r from the consensus is above the sample median, and equals zero otherwise. The consensus is the prevailing consensus on the day prior to the forecast, and is calculated as the weighted average, by the age of the forecast, of all analysts' latest forecasts for firm i for fiscal quarter t .
$Effort_r$	An indicator variable that equals one if the number of firms covered by the analyst issuing forecast r during quarter t is below the sample median, and equals zero otherwise.
$InfoN_{i,d}^{IR}$	An indicator variable that equals one if trading day d is the information release announcement date ($N=0$) or the trading day subsequent to the announcement date ($N=1$) and equals zero otherwise. IR can refer to an analyst forecast ($IR=AF$), earnings announcement ($IR=EA$), or management forecast ($IR=MF$).
$Last_r^{AF}$	In a sub-sample of the initial forecasts issued by each analyst for firm i during quarter t , an indicator variable that equals one when forecast r is the latest such forecast during the quarter, and equals zero when forecast r is the earliest such forecast during the quarter.
$LnSize_{i,t}$	Log of firm i 's average market value of equity during quarter t .
$PostInfoN_{i,d}^{IR}$	An indicator variable that equals one if trading day d is the N^{th} trading day after the announcement day of an information release, and equals zero otherwise. IR can refer to an analyst forecast ($IR=AF$), earnings announcement ($IR=EA$), or management forecast ($IR=MF$). N ranges from 2 to 10.
$PreInfoN_{i,d}^{IR}$	An indicator variable that equals one if trading day d is the N^{th} trading day preceding the announcement day of an information release, and equals zero otherwise. IR can refer to an analyst forecast ($IR=AF$), earnings announcement ($IR=EA$), or management forecast ($IR=MF$). N ranges from 1 to 10.
$Price_{i,d}$	Firm i 's stock price on day d , obtained from the daily CRSP file.
$PriceImpact_{i,d}$	The average transaction-based dollar price impact for all trades in firm i 's stock on day d (multiplied by 100). The price impact of the k^{th} trade is computed as $2 * M_{k+5} - M_k $, where M_{k+5} is the midpoint of the bid and offer five minutes after the midpoint M_k (Holden and Jacobsen, 2014).
$Revision_r$	An indicator variable that equals one if the absolute percentage revision in analyst forecast r is above the sample median, and equals zero otherwise. We calculate percentage revision as the current forecast for firm i minus the previous forecast by the same analyst for firm i , scaled by that previous forecast. We further require both forecasts to be made after the previous quarter's earnings announcement.
$Turnover_{i,t}$	Average daily stock turnover of firm i during quarter t .
$Volatility_{i,t}$	Firm return volatility, defined as the standard deviation of firm i 's daily stock return during fiscal quarter t .
$Volume_{i,d}$	Firm i 's trading share volume on day d , obtained from the daily CRSP file (divided by 1,000,000).

Note: Subscripts r , i , t , and d refer to information release r , firm i , fiscal quarter t , and trading day d .

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